

# Vigneshraja Ganeshraj

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## Executive Summary

Highly motivated MSc chemist (graduating May 2019) with 3 years of exceptional analytical research experience. Skilled in several analytical techniques including publication quality characterization of a wide range of organic and inorganic compounds in high resolution ESI-MS, MALDI-MS, optical spectroscopy including UV-Vis, and x-ray crystallography. Additionally, experienced in several other analytical techniques and skills, including microscopy, HPLC-MS, GC-MS, ICP-MS and 3 years of experience in size exclusion chromatography. A demonstrated history of excellent communication through the presentation of research to a broad range of audiences, and 2 publications in peer-reviewed journals. Detail oriented leader, with supervisory experience, and a commitment to excellence. Enjoy working in both a team or autonomous environment.

|                  |                                                                                                                                                                                              |                 |
|------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|
| <b>Education</b> | <b>Master of Science in Chemistry</b><br><i>The University of Mississippi, Oxford, MS</i><br>Research Advisor: Dr. Amala Dass                                                                | <i>May 2019</i> |
|                  | <b>Bachelor of Technology in Chemical &amp; Electrochemical Engineering</b><br><i>Central Electrochemical Research Institute, Karaikudi, INDIA</i><br>Research Advisor: Dr. S. Muthukrishnan | <i>May 2014</i> |

## Professional Experience

- *Teaching Assistant* *Jan 2019-Present*
  - The University of Mississippi, Oxford, MS; Supervisor Dr. John Wiginton
- *Lab Technician (Internship)* *Dec 2018-Jan 2019*
  - High End Multi Processing LLC, Spencer, NY
    - Initial Extraction of cannabinoids from Biomass using CO<sub>2</sub> extractor and Ethanol extraction techniques.
    - Separation of cannabinoids majorly CBD (~80% pure) using distillation process.
    - Isolation of CBD (~99% pure) using flash chromatography. Trained from BUCHI field service engineer to perform analysis using Reveleris X2- C18 column.
    - Analysis of samples using HPLC and quality control of the full spectrum cannabinoid oil.
- *Research Assistant* *Aug 2018-Dec 2018*
  - The University of Mississippi, Oxford, MS; Supervisor Dr. Amala Dass
    - Experienced in several analytical techniques including: nESI-Q-TOF-MS, MALDI-TOF-MS, UV-Vis spectroscopy, low temperature UV-Vis spectroscopy, x-ray crystallography (sc-XRD), ICP-MS, HPLC-MS, Size exclusion chromatography, and electrochemical methods (CV and DPV)
    - Maintenance, troubleshoot and repair of analytical instruments
- *Mass Spectrometry Analyst* *Aug 2017-Aug 2018*
  - The University of Mississippi, Oxford, MS; Supervisor Dr. Amala Dass
    - Analysis of the samples using ESI and MALDI mass spectrometry for publication quality determination of a wide variety of compounds for the Department of Chemistry & Biochemistry and Department of Pharmacy
- *Teaching Assistant* *Aug 2016-Aug 2017*
  - The University of Mississippi, Oxford, MS; Supervisor Dr. John Wiginton

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- *Process Engineer*

*Jan 2016-May 2016*

- Tes-Amm India Pvt. Ltd, Chennai, India

- Secondary metal recovery of gold (Au) and silver (Ag) from electronic wastes using Acid digestion and Electroplating technique.
    - Secondary metal recovery of Palladium (Pd) from catalytic converter of automobiles using Acid digestion technique.

- *Research Analyst*

*Jun 2014-Dec 2015*

- Shanpoornam Metals Sdn.Bhd, Penang, Malaysia

- Secondary recovery of gold from LED light manufacturing wastes (majorly from OSRAM) using Acid digestion technique (Hydrometallurgy).
    - Secondary recovery of platinum (Pt) metal from hard disks using Electrodeposition technique.
    - Step by step process monitoring and analysis of the samples using ICP-MS, XRF and AAS.
    - Production planning and execution.

### Research Experience

**Graduate Research Experiences in the laboratory of Dr. Amala Dass,  
Department of Chemistry & Biochemistry, The University of Mississippi.  
2 publications and 3 graduate presentations**

*Jul2016- Present*

### Technical Skills

- nESI-Q-TOF-MS high resolution electrospray ionization mass spectrometry
- MALDI-TOF-MS matrix assisted laser desorption ionization mass spectrometry
- UV-Vis-NIR spectroscopy
- Single Crystal x-ray diffraction (crystallography)
- HPLC-MS (high performance liquid chromatography)
- Flash Chromatography (BUCHI Reveleris X2)
- Size exclusion chromatography using 6'' to 6' in length columns, using milligram to gram quantities
- Proficient with Endnote software for reference and citation management
- Proficient in Microsoft Word, Excel, and PowerPoint

### References

Dr. Walter E. Cleland, Associate Chair and Associate Professor of Chemistry & Biochemistry, University of Mississippi

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*\*Additional References available upon request*